FT-02 V2: 9V PD preferred / conservative 5V fallback / one NVDC / physical system OFF

CHARGE INHIBITED. ERC is not a safety certificate. Thermal, PD NVM and battery gates remain OPEN.

01_USB_PD

文件: 01_USB_PD.kicad_sch

02_INPUT_PROTECTION

文件: 02_INPUT_PROTECTION.kicad_sch

03_CHARGER_NVDC

文件: 03_CHARGER_NVDC.kicad_sch

04_BATTERY_GAUGE

文件: 04_BATTERY_GAUGE.kicad_sch

05_HARD_OFF

文件: 05_HARD_OFF.kicad_sch

06_3V3_RAIL

文件: 06_3V3_RAIL.kicad_sch

07_5V_RAIL

文件: 07_5V_RAIL.kicad_sch

08_HOST_INTERFACE

文件: 08_HOST_INTERFACE.kicad_sch

REVIEW DRAFT – CHARGE INHIBITED – NO PCB RELEASE

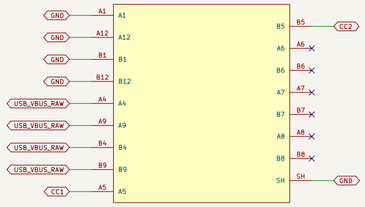
Sheet: /
File: FT02_PowerModule_V2.kicad_sch

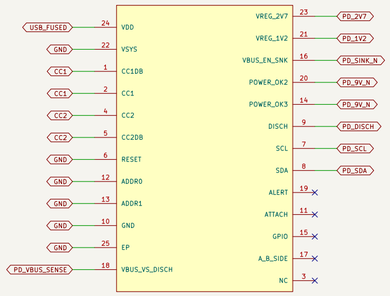
Title: FT02_PowerModule_V2

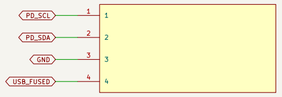
Size: A2	Date:	**Rev: V2-PD-R0**
KiCad E.D.A. 10.0.5		Id: 1/9

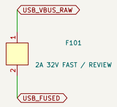
STUSB4500: program and VERIFY PDO1=5V/0.1A, PDO2=9V/2A, PDO3=9V/3A before use.

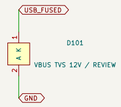
Local symbols use datasheet pin identities. Footprints/connector pin-to-pad maps are NOT released.

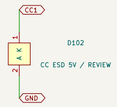


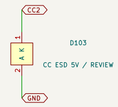


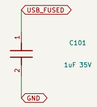






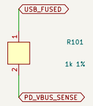


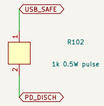


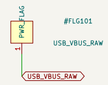


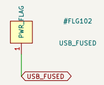














Use review/SCHEMATIC_REVIEW.md for source, charge lock, PWR_FLAG and netlist proof boundaries.

REVIEW DRAFT — CHARGE INHIBITED — NO PCB RELEASE

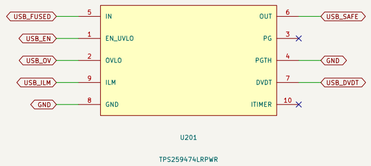
Sheet: /01_USB_PD/
File: 01_USB_PD.kicad_sch

Title: 01_USB_PD

Size: A2	Date:	**Rev: V2-PD-R0**
KiCad E.D.A. 10.0.5		Id: 2/9

One input eFuse, not a source MUX. Independent OV cutoff also blocks factory-default 15V/20V PDOs.

Local symbols use datasheet pin identities. Footprints/connector pin-to-pad maps are NOT released.



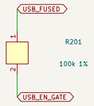
U201

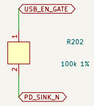
TPS2594744LRPWR

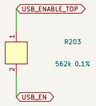


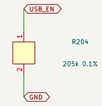
Q201

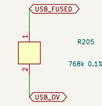
BSS84 signal only

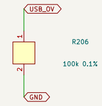


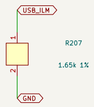
















Use review/SCHEMATIC_REVIEW.md for source, charge lock, PWR_FLAG and netlist proof boundaries.

REVIEW DRAFT — CHARGE INHIBITED — NO PCB RELEASE

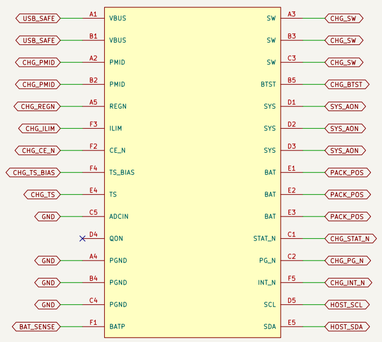
Sheet: /02_INPUT_PROTECTION/
File: 02_INPUT_PROTECTION.kicad_sch

Title: 02_INPUT_PROTECTION

Size: A2	Date:	**Rev: V2-PD-R0**
KiCad E.D.A. 10.0.5		Id: 3/9

BQ25638 SYS is the main source. CE arm DNP + TS forced fault: CHARGING IS PHYSICALLY INHIBITED.

Local symbols use datasheet pin identities. Footprints/connector pin-to-pad maps are NOT released.



U301

BQ25638YBGR



Q301

MMBT3906



Q302

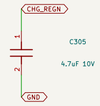
2N7002

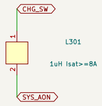


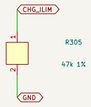


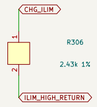


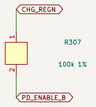




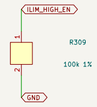


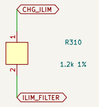






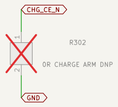












Use review/SCHEMATIC_REVIEW.md for source, charge lock, PWR_FLAG and netlist proof boundaries.

REVIEW DRAFT — CHARGE INHIBITED — NO PCB RELEASE

Sheet: /03_CHARGER_NVDC/
File: 03_CHARGER_NVDC.kicad_sch

Title: 03_CHARGER_NVDC

Size: A2	Date:	**Rev: V2-PD-R0**
KiCad E.D.A. 10.0.5		Id: 4/9

Protected pack only. PACK_NEG is isolated from board GND except the four-terminal shunt.

Local symbols use datasheet pin identities. Footprints/connector pin-to-pad maps are NOT released.



J401

PROTECTED 1S PACK



R401

5m 1% >=0.5W 4T



U401

MAX17055ETB+T









Use review/SCHEMATIC_REVIEW.md for source, charge lock, PWR_FLAG and netlist proof boundaries.

REVIEW DRAFT — CHARGE INHIBITED — NO PCB RELEASE

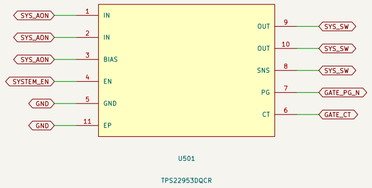
Sheet: /04_BATTERY_GAUGE/
File: 04_BATTERY_GAUGE.kicad_sch

Title: 04_BATTERY_GAUGE

Size: A2	Date:	**Rev: V2-PD-R0**
KiCad E.D.A. 10.0.5		Id: 5/9

Physical switch OFF grounds gate EN. Charger and gauge are upstream. UV restart/thermal tests remain OPEN.

Local symbols use datasheet pin identities. Footprints/connector pin-to-pad maps are NOT released.



U501

TPS22953DQCR



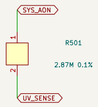
U502

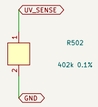
TPS3808G01DBVR

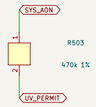


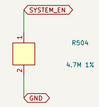
SW501

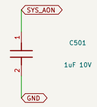
SYSTEM ON / OFF







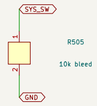


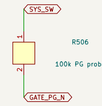












Use review/SCHEMATIC_REVIEW.md for source, charge lock, PWR_FLAG and netlist proof boundaries.

REVIEW DRAFT — CHARGE INHIBITED — NO PCB RELEASE

Sheet: /05_HARD_OFF/
File: 05_HARD_OFF.kicad_sch

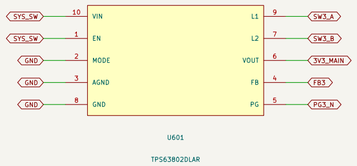
Title: 05_HARD_OFF

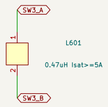
Size: A2 | Date: | **Rev: V2-PD-R0**

KiCad E.D.A. 10.0.5 | Id: 6/9

TPS63802 3V3/2A individual target; aggregate output budget 8W, not both rails at maximum.

Local symbols use datasheet pin identities. Footprints/connector pin-to-pad maps are NOT released.

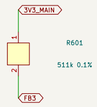


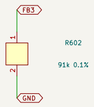


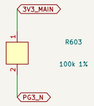


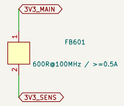






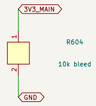












Use review/SCHEMATIC_REVIEW.md for source, charge lock, PWR_FLAG and netlist proof boundaries.

REVIEW DRAFT — CHARGE INHIBITED — NO PCB RELEASE

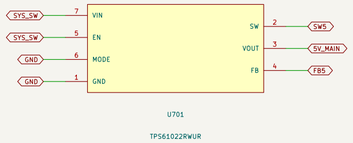
Sheet: /06_3V3_RAIL/
File: 06_3V3_RAIL.kicad_sch

Title: 06_3V3_RAIL

Size: A2	Date:	**Rev: V2-PD-R0**
KiCad E.D.A. 10.0.5		Id: 7/9

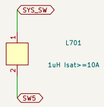
TPS61022 BOOST, not buck; 5V/1.5A individual target. No rail derives directly from USB VBUS.

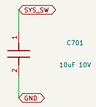
Local symbols use datasheet pin identities. Footprints/connector pin-to-pad maps are NOT released.

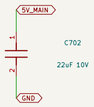


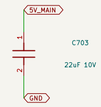
U701

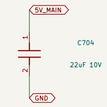
TPS61022RWUR

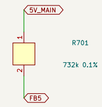


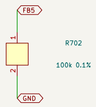




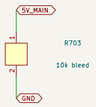












Use review/SCHEMATIC_REVIEW.md for source, charge lock, PWR_FLAG and netlist proof boundaries.

REVIEW DRAFT — CHARGE INHIBITED — NO PCB RELEASE

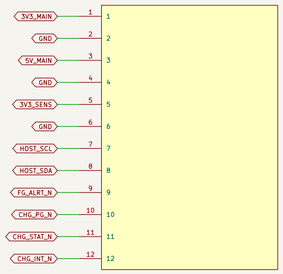
Sheet: /07_5V_RAIL/
File: 07_5V_RAIL.kicad_sch

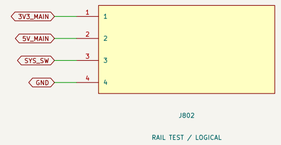
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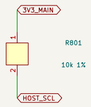
Size: A2	Date:	**Rev: V2-PD-R0**
KiCad E.D.A. 10.0.5		Id: 8/9

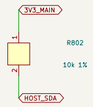
All host pullups use switched 3V3. Private PD service bus must NOT join the host bus.

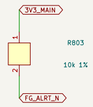
Local symbols use datasheet pin identities. Footprints/connector pin-to-pad maps are NOT released.

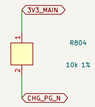


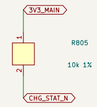


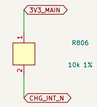












Use review/SCHEMATIC_REVIEW.md for source, charge lock, PWR_FLAG and netlist proof boundaries.

REVIEW DRAFT — CHARGE INHIBITED — NO PCB RELEASE

Sheet: /08_HOST_INTERFACE/
File: 08_HOST_INTERFACE.kicad_sch

Title: 08_HOST_INTERFACE

Size: A2	Date:	**Rev: V2-PD-R0**
KiCad E.D.A. 10.0.5		Id: 9/9